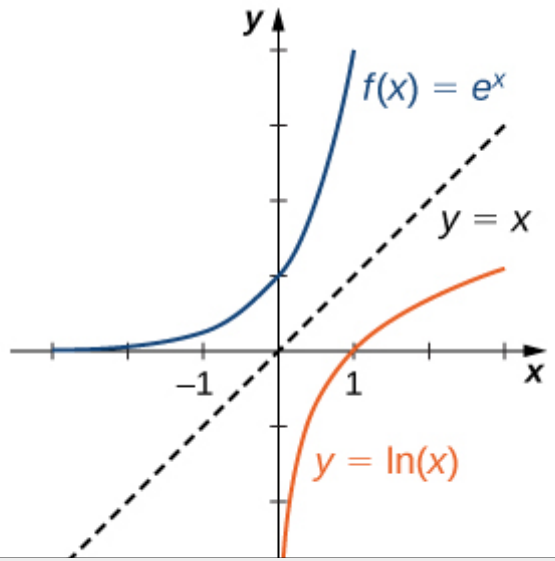
2627-MA140 Engineering Calculus

Week 05, Lecture 3
Exponentials and Logarithms; Higher-order derivatives

Dr Niall Madden

University of Galway

Thursday, 8 October, 2026



Naturally, today we shall discuss...

1. Exponential functions
2. Properties of exponents
3. The number e
4. Natural Exponential Function
   - The derivative of e^x
5. Logarithms
   - Properties
   - The natural logarithm
   - Derivative of $\ln(x)$
6. Logarithmic differentiation
7. Higher-order Derivatives
8. Exercises

Slides are on canvas, and at niallmadden.ie/2627-MA140



See also: 3.9 (Derivatives of Exponential and Logarithmic Functions) of **Calculus** by Strang & Herman: https://math.libretexts.org/Bookshelves/Calculus/Calculus_(OpenStax)

Exponential functions

Earlier in this course we met functions such as $y = x^2$; this is a **power** function.

Now we consider **exponential functions**, such as $y = 2^x$.

Such functions occur in many applications. For example: if I invest €100 with an annual interest rate of 20%, then after x years, I will have €$100 \times (1.2)^x$. **Why?**

Exponential functions

Exponential functions grow quite fast: if my investment is indeed worth $f(x) = 100 \times (1.2)^x$ euros after x years, then...

- After 1 year, I have €120
- After 10 years, I have €619.17
- After 20 years, I have €3,833.80
- After 25 years, I have €9,539.60
- After 50 years, and 190 days, I'll be a millionaire!

Properties of exponents

Here I remind you of some properties of exponents that you should already know: for any positive numbers a and b, and for all x and y,

1. $b^x b^y = b^{x+y}$
2. $\dfrac{b^x}{b^y} = b^{x-y}$
3. $(b^x)^y = b^{xy}$
4. $(ab)^x = a^x b^x$
5. $\left(\dfrac{a}{b}\right)^x = \dfrac{a^x}{b^x}$

There is more information on this in Section 1.5 of the textbook (Exponential and Logarithmic Functions). ($\longleftarrow$ link to book)

The number e

The number e is a mathematical constant (similar, in a sense, to the way that $\pi \approx 3.14159$ is too).

The value of e is roughly 2.7182818284.

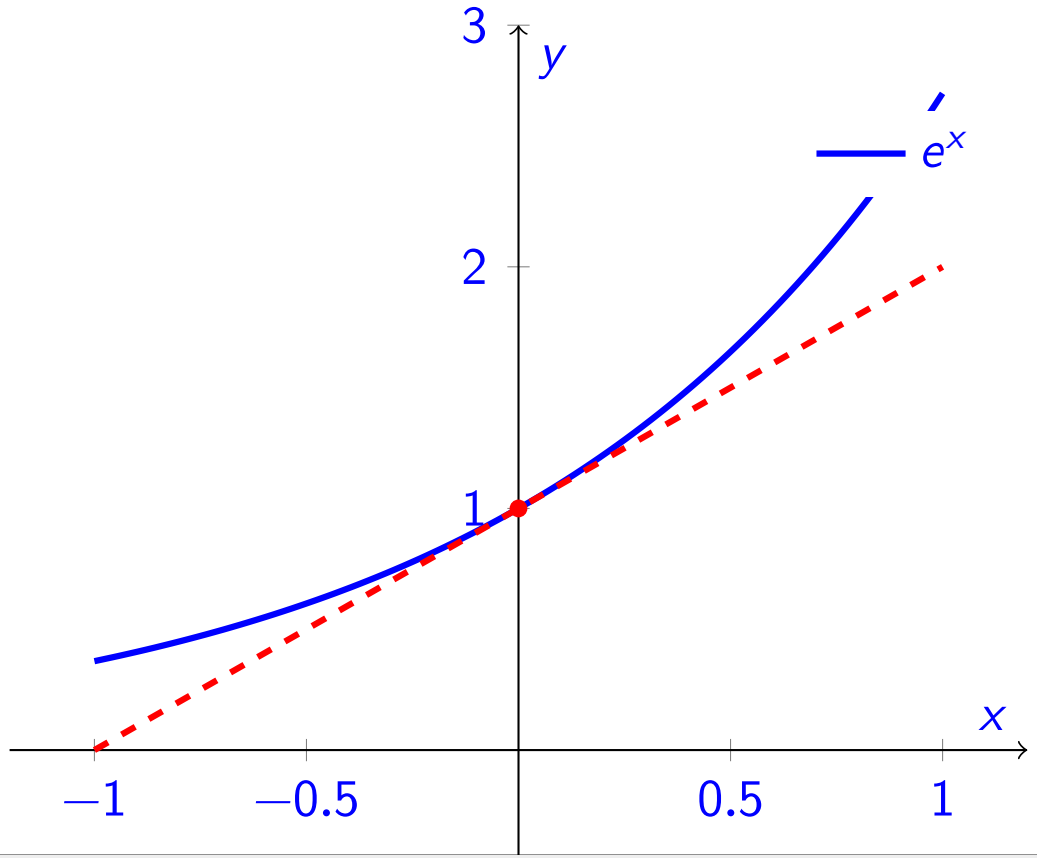
It is often called **Euler's Number** after Leonard Euler, who did not discover it: that was (probably) Jacob Bernoulli in 1683 while studying compound interest. Or maybe 100 years earlier by John Napier.

It has some very interesting and important properties...

Natural Exponential Function

The Natural Exponential Function

The Natural Exponential Function is $f(x) = e^x$. It is special for many reasons, including the its tangent at $x = 0$ has slope 1.



Let's assume that e is the number for which, if $f(x) = e^x$, then $f'(0) = 1$. Using the limit definition of the derivative, this means

$$f'(0) = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \frac{e^{0+h} - e^0}{h} = \lim_{h\to 0} \frac{e^h - 1}{h} = 1.$$

Now we use this to work out $\dfrac{d}{dx}[e^x]$.

So now we know that

$$\frac{d}{dx}e^x = e^x.$$

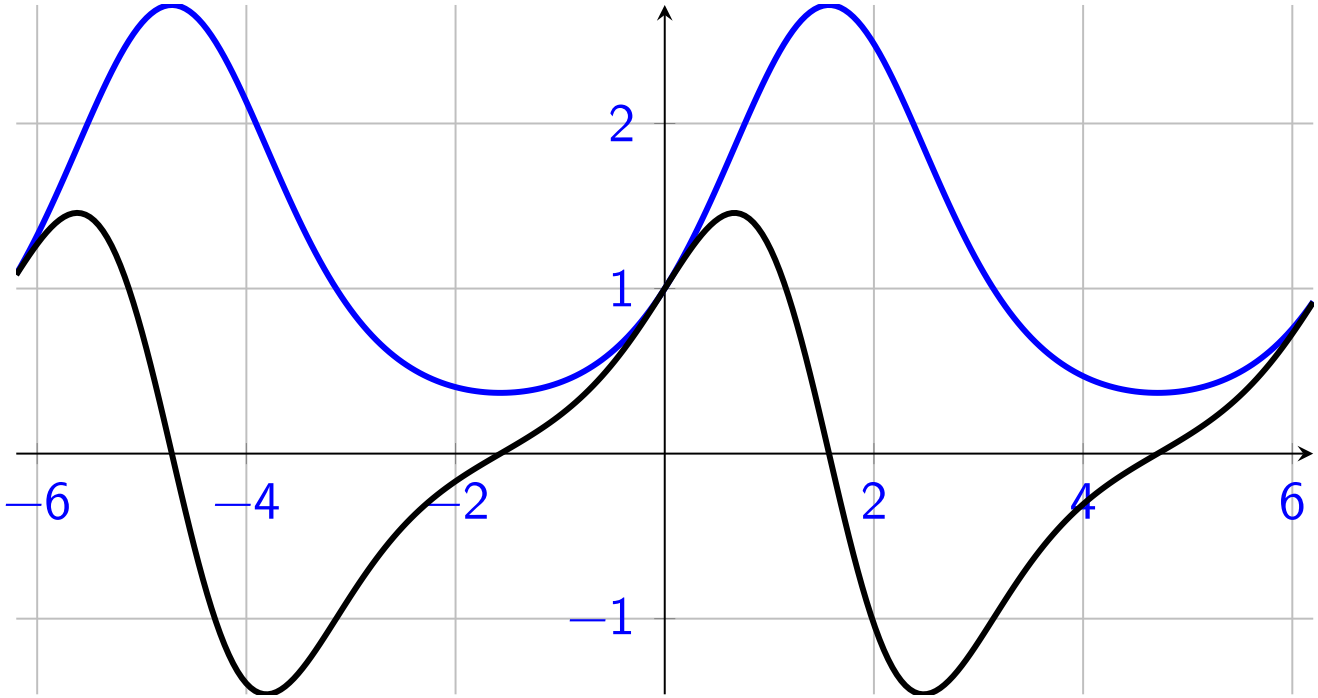
That is e^x is the function that is its own derivative!!!

Example

Compute the derivative of $f(x) = e^{\sin(x)}$

Graphs of $f(x)$ and $f'(x)$ from the previous example.

Which is which?



Logarithms

Suppose that $y = f(x)$ is an **exponential** function; that is: $y = b^x$ for some $b > 0$ (and excluding $x = 1$).

Its **inverse** is called a **logarithmic function**, denoted $\log_b$

$$\text{If } y = b^x \quad \text{then} \quad \log_b(y) = x.$$

Examples

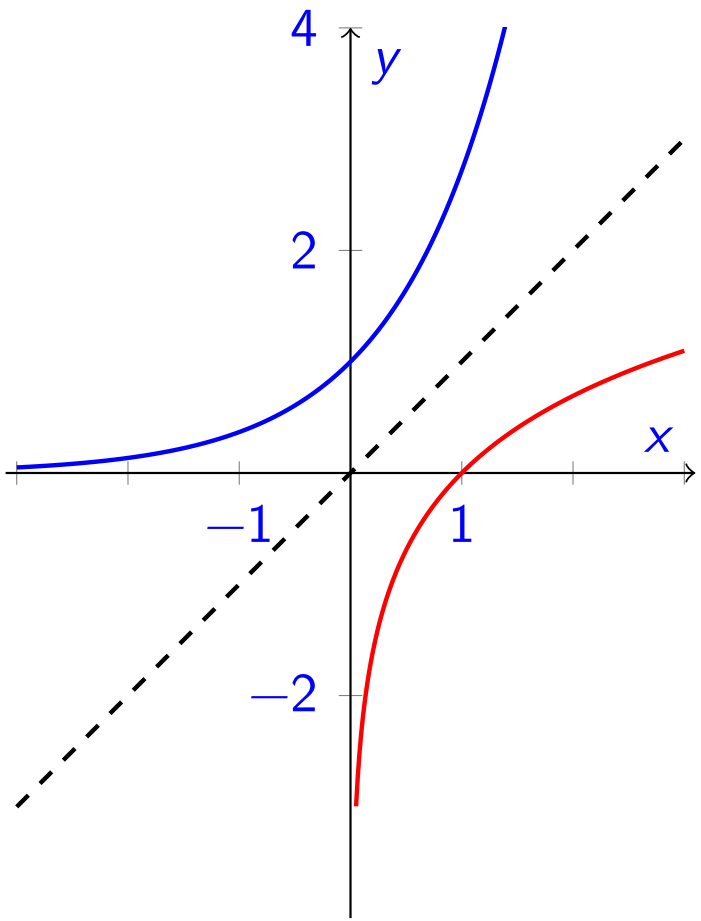
- $\log_2(8) = 3$
- $\log_{10}(100) = 2$
- $\log_e(e^x) = x$

Properties of Logarithms

If $a, b, c > 0$ and $b \neq 1$, then

- $\log_b(ac) = \log_b(a) + \log_b(c)$
- $\log_b\left(\frac{a}{c}\right) = \log_b(a) - \log_b(c)$
- $\log_b(a^r) = r \log_b(a)$

We denote $\log_e(x)$ as $\ln(x)$



$$\frac{d}{dx}\ln(x) = \frac{1}{x}$$

Why?

Example:

Find the derivative of $f(x) = \ln(x^2 + 2x + 3)$.

Logarithmic differentiation

We can use logs when computing derivatives: **Logarithmic differentiation**, which helps us differentiate functions with x, or a function of x in the exponent, such as $y = (2x)^{\sin(x)}$ or $y = x^x$.

Strategy:

- Take $\ln()$ of both sides
- Simplify, using properties of logarithms.
- Differentiate.
- Solve for $\dfrac{dy}{dx}$

Logarithmic differentiation

Example

Differentiate $f(x) = x^x$.

Logarithmic differentiation

Higher-order Derivatives

We learned last week that the derivative of $f(x)$, denoted $f'(x)$, is itself a function.

That implies that $f'(x)$ can itself be differentiated, which is called the **second derivative** of f. It is denoted as

$$\frac{d^2y}{dx^2} \quad \text{or} \quad f''(x) \quad \text{or} \quad f^{(2)}(x).$$

We can continue this process to get higher-order derivatives as long as the preceding derivative is again differentiable.

The first and second derivatives f' and f'' (if they exist) provide valuable information about the function and its graph, particularly concerning local or global maxima, local/global minima and points of inflection.

Higher-order Derivatives

Example

Find the **second** derivative of the functions

(i) $f_1(x) = 3x^2 + 2x + 1$

(ii) $f_2(x) = e^x$

(iii) $f_3(x) = \ln x$

(iv) $f_4(x) = \sin(x)$

Exercises

Exercise 5.3.1

(a) Show that if $f(x) = e^{Cx}$ for some constant C, then $f'(x) = Ce^{Cx}$.

(b) If $f(x) = e^{Cx}$, find all values of C such that $f''(x) = f(x)$.

Exercise 5.3.2 [2025 exam, Q4]

(a) Differentiate $f(x) = x^2 e^{-3x} \sin(x/4)$ with respect to x.

(b) Differentiate $f(\theta) = \dfrac{\sin(3\theta) + 1}{2\theta + 5}$ with respect to θ.

(c) Differentiate $f(x) = \ln\left(\cos(x^2)\right)$ with respect to x.